

# PURUSHOTTAM KUMAR

SOC & BLUE TEAM · SECURITY ANALYST INTERN

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## SUMMARY

Second-year B.Tech Computer Science (Cyber Security) student seeking a SOC or security analyst internship. Completed the **TryHackMe SOC Level 1** path — 65 hours of security monitoring, log and network analysis, threat intelligence and incident response — working hands-on in **Wireshark**, **Splunk** and **Windows event logs**. Build detection tooling in Python: a live-traffic intrusion detection system running signature and machine-learning engines in parallel and emitting SIEM-ready JSON alerts. Currently preparing for CompTIA Security+.

## EDUCATION

**Central University of Jammu**  
*B.Tech, Computer Science & Engineering — Cyber Security Specialization*  
2nd year · expected graduation 2029  
**Relevant coursework:** Computer Networks, Data Structures & Algorithms, Cyber Security Fundamentals, Machine Learning

2025 – 2029  
CGPA: 8.12 / 10

## TECHNICAL SKILLS

|                     |                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOC & Detection     | Security monitoring and alert triage, log analysis, Splunk (TryHackMe labs), signature- and anomaly-based intrusion detection, incident response workflow, threat intelligence     |
| Network Analysis    | Wireshark, packet capture & analysis (Scapy), TCP/IP, socket programming, port scanning & banner grabbing, VLANs and trunking, routing protocols, ACLs (Cisco Packet Tracer, GNS3) |
| Endpoints & Systems | Windows Event Log, Sysmon, Linux administration and log analysis, endpoint security                                                                                                |
| Languages           | Python, C, C++, JavaScript                                                                                                                                                         |
| Tools & Platforms   | Git & GitHub, Scikit-learn, React, Cloudflare Workers, Vercel, TryHackMe                                                                                                           |

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Network Intrusion Detection System</b> — <a href="#">source ↗</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026 |
| Python · Scapy · Scikit-learn · Isolation Forest · Multithreading                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |
| <ul style="list-style-type: none"><li>Built a real-time IDS that sniffs live traffic with Scapy and runs two detection engines in parallel: signature rules for SYN floods and port scans, and an Isolation Forest model that flags anomalous, previously unseen traffic.</li><li>Emits structured JSON alerts — timestamp, threat type, source and destination IP, confidence score — <b>formatted for direct SIEM ingestion</b>; multithreaded capture-and-analysis pipeline verified by an automated test suite passing 4/4 cases.</li></ul> |      |
| <b>Threaded TCP Port Scanner</b> — <a href="#">source ↗</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2025 |
| Python · Sockets · Multithreading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |
| <ul style="list-style-type: none"><li>Wrote a multithreaded scanner with service banner grabbing and per-socket timeout handling, enumerating open TCP ports and running services for fast, low-noise attack-surface mapping.</li></ul>                                                                                                                                                                                                                                                                                                         |      |
| <b>FTP Server From Scratch</b> — <a href="#">source ↗</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025 |
| Python · Raw Sockets · TCP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |
| <ul style="list-style-type: none"><li>Implemented an FTP server on raw sockets covering core file-transfer operations, written without frameworks to work through client-server protocol internals firsthand.</li></ul>                                                                                                                                                                                                                                                                                                                         |      |
| <b>Pluton — AI Code Reviewer</b> — <a href="#">source ↗</a> · <a href="#">live demo ↗</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026 |
| TypeScript · React · Cloudflare Workers · Gemini API                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |
| <ul style="list-style-type: none"><li>Shipped a web app that auto-detects a submitted file's language and returns a scored static review, flagging insecure patterns alongside performance, readability and maintainability findings; deployed serverless on Cloudflare Workers at the edge.</li></ul>                                                                                                                                                                                                                                          |      |

## CERTIFICATIONS

**TryHackMe SOC Level 1** — 65 hrs, completed 2026 · security monitoring, log and network analysis, threat intelligence, SIEM and incident response · [verify ↗](#)  
**CompTIA Security+** — currently preparing  
**Cisco Networking Academy** — Endpoint Security · Networking Devices & Basic Configuration · Networking Basics (2026)

## ACHIEVEMENTS & ACTIVITIES

- Snyk CTF** — **top 22%**, placing 328 of 1,500 competing teams.
- Active CTF competitor on TraceBash, working forensics, steganography, binary exploitation, OSINT and web challenges.
- Six projects shipped and source-linked on [GitHub](#), catalogued with live demos on a [self-built portfolio site](#).